Narwan Joyenda

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EDUCATION

The George Washington University, Washington, DC — Aug 2024 – May 2028
Bachelor of Science in Computer Engineering,
Honors: U.S. Presidential Scholar
Relevant Coursework: Data Structures & Algorithms, Design of Logic Systems, Microprocessor Systems, Computer Organization, Digital Signal Processing, Pattern Recognition & Machine Learning

TECHNICAL SKILLS

Programming & HDL: Python, C, MATLAB, Verilog
AI & Automation: OpenAI API, n8n, Generative AI, AI Agents, Prompt Engineering, Prompt Evaluation, Model Output Validation
Cloud & Development Tools: AWS EC2, S3, IAM, Git/GitHub, VS Code, Jupyter Notebook, GitHub Pages
EDA, CAD & Simulation Tools: Tanner S-Edit, Multisim, LTspice, Tinkercad, AutoCAD, SolidWorks

TECHNICAL PROJECTS

Audio Amplifier Speaker System – George Washington University (ECE 2115) — May 2026

- Designed and assembled an LM386-based analog audio amplifier with bass and treble filtering, integrating discrete components and signal-conditioning circuitry into a functional prototype.
- Characterized frequency response, voltage gain, and output behavior using an oscilloscope and function generator; troubleshot the circuit to reduce distortion and improve audio quality.

AI Agent Development – George Washington University (ECE 1020) — Apr 2025

- Built and deployed an interactive AI agent using the OpenAI API, n8n, and GitHub Pages, integrating API workflows, user-input processing, tool-based memory, and real-time responses.
- Tested and debugged the application workflow to improve response handling, reliability, and the overall user experience

EXPERIENCE

George Washington University – HPCAT Laboratory, Washington, DC — Sep 2026 – Present
Undergraduate Research Assistant

- Conduct research in computer architecture and high-performance computing, focusing on energy-efficient, reliable, scalable, and secure parallel systems for AI and machine learning workloads.
- Investigate NoC, manycore, and accelerator-based architectures, including chiplet-based ML accelerators, hardware-software co-design, and emerging photonic/interconnect technologies for high-performance, sustainable computing.

George Washington University, Washington, DC — Jun 2026 – Aug 2026
Summer Undergraduate Research Assistant – LLM Reasoning & Ill-Posed Problem Generation

- Built and validated a dataset of 500+ ill-posed quantitative reasoning problems using K=3 and K=4 branching structures, developing Python-based generation logic to create multiple resolution paths, conflicting conditions, and hidden constraints while maintaining mathematical consistency.
- Evaluated LLM responses for reasoning errors, branch inconsistencies, and unsupported conclusions, identifying recurring failure patterns and refining the dataset validation process.

GAP Inc. — Sales & Operations Assistant, Tysons, VA — Jul 2023 – Aug 2024
Meta Marketing & Consulting Services, Remote — Sep 2022 - Mar 2023
IT Intern

- Supported software, hardware, firewall, and user-authentication troubleshooting while tracking technical requests and documenting recurring issues and resolutions.

LEADERSHIP & ACTIVITIES

AWS Student Builder Club – George Washington University, Washington, DC — Dec 2025 – Present
Programming Director & Technical Lead

- Lead technical programming and support hands-on workshops covering AWS EC2, S3, IAM, cloud architecture, Git/GitHub, and application deployment.

GW Robotics Team – George Washington University, Washington, DC — Sep 2026 – Present
Secretary

- Support team organization, meeting coordination, documentation, and communication for club activities and engineering projects.

IEEE – George Washington University, Washington, DC — Jan 2025 – Present